

Answer all questions

- 1 (a) The equation of a curve is $y = 2x^2 + ax + (6 + a)$, where a is a constant. Find the range of values of a for which the curve lies completely above the x -axis. [3]

- (b) The equation of a curve is $y = 3x^2 + 4x + 6$.
(i) Find the set of values of x for which the curve is above the line $y = 6$. [3]

- (ii) Show that the line $y = -8x - 6$ is a tangent to the curve. [2]

2 Express $\frac{4x^2 + 6x + 5}{2x^2 + x - 3}$ in partial fractions.

[5]

3 It is given that $y = 1 - \frac{1}{2}\sin 3x$, $0^\circ \leq x \leq 240^\circ$.

(i) State the maximum and minimum values of y .

[2]

(ii) Sketch the graph of $y = 1 - \frac{1}{2}\sin 3x$ for $0^\circ \leq x \leq 240^\circ$.

[3]

4 It is given that $y = \frac{2x^2}{4x-3}$, where $x > \frac{3}{4}$.

(i) Find $\frac{dy}{dx}$.

[2]

(ii) Find the range of values of x for which $y = \frac{2x^2}{4x-3}$ is a decreasing function.

[4]

5 It is given that $f(x)$ is such that $f'(x) = \frac{1}{2x-5} - \frac{4}{(2x-5)^2}$.

Given also that $f(3) = 1.75$, show that $8f(x) - (2x-5)^2 f''(x) = \ln(2x-5)^4$.

[7]

6 (i) Show that $\cot 2x = \frac{1 - \tan^2 x}{2 \tan x}$. [2]

(ii) Hence, solve the equation $8 \cot 2x \tan x = 1$, for $0^\circ < x < 360^\circ$. [4]

7 The function $f(x)$ is such that $f(x) = 2x^3 + 3x^2 - x - 4$.

(i) Find a factor of $f(x)$.

[2]

(ii) Hence, determine the number of solutions of the equation $f(x) = 0$.

[4]

8 It is given that $f(x) = -2x^3 + 5x^2 + 4x + a$.

(i) Find the coordinates of the turning points in terms of a . [4]

(ii) Determine the nature of each turning point. [3]

(iii) In the case where $a = 1$, explain why the part of the graph between the turning points lie above the x – axis. [2]

9 (i) Show that $\sec x + \tan x$ can be expressed as $\frac{1 + \sin x}{\cos x}$. [2]

(ii) Differentiate $\ln(\sec x + \tan x)$ with respect to x . [3]

(iii) Hence, find $\int_{0.25}^{0.5} 2 \sec x \, dx$. [3]

- 10** A ball is thrown vertically upwards. Its height, h m, above the ground at time t seconds after being thrown is given by the formula $h = -4.9t^2 + 20t + 1.5$.

(a) State the height above the ground from which the ball is thrown. [2]

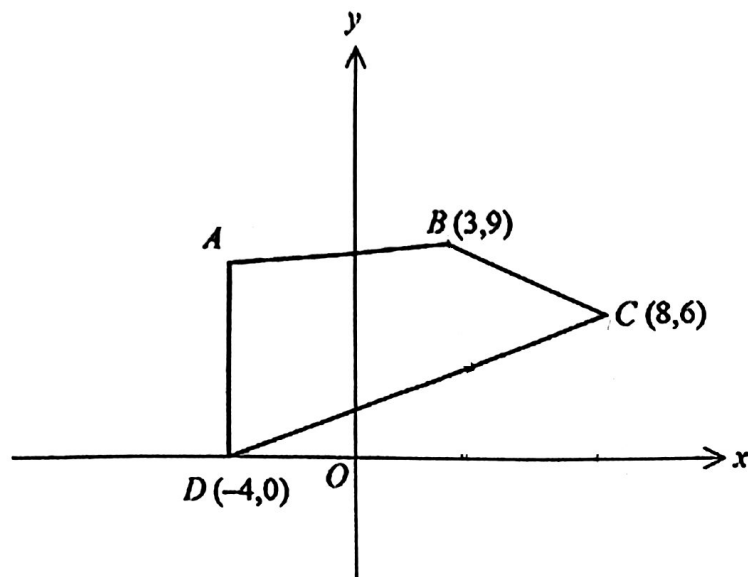
(b) Express h in the form $a + b(t + c)^2$ where a , b and c are constants to be determined. [3]

(c) Hence, state the maximum height attained by the ball and the time which this occurs. [2]

- (d) The ball hits the ground. Explain why the time taken for the ball to hit the ground is not twice the time found in part (c). [1]

- (e) Find the length of time for which the ball is at least 2 m above the ground. [3]

11



A quadrilateral $ABCD$ passes through vertices $B(3, 9)$, $C(8, 6)$ and $D(-4, 0)$, line AD is parallel to the y – axis.

(i) Find the coordinates of A given that the length of AD is 8 units. [1]

(ii) A point P divides the line DC in the ratio of $2 : 1$. Find the coordinates of P . [2]

(iii) Hence, find the area of the quadrilateral $ABPD$. [3]

- 12** The points A and B lie on the circumference of a circle C_1 where A is the point $(0, 8)$ and B is the point $(4, 0)$. The line $y = 2x$ also passes through the centre of the circle C_1 .

(i) Find the centre and radius of the circle C_1 .

[5]

- (ii) Find the equation of the circle C_1 in the form $x^2 + y^2 + px + qy + r = 0$, where p, q and r are integers.

[2]

Another circle C_2 of radius $\sqrt{2}$ units has its centre inside C_1 and it cuts the circle C_1 at the origin and at the point on the circumference where $x = 2$.

- (iii) Find the centre of C_2 .

[6]

Continuation of working space for Question 12.

- 1 The population of turtles in a lake is modelled by $P = 3250 + Ae^{-kt}$, where P is the number of turtles t years after monitoring began. It is given that initially there were 4400 turtles, and after 10 years there are 3900.

(i) Find the values of A and k .

[3]

(ii) Derive, with clear working, the population of turtles in the long run.

[2]

- 2 (a) Given that $\left(25\sqrt{7^3}\right)^{-2} \div \left(\frac{49}{125}\right)^{1-q} = 5^p$, find the values of p and q .

(b) It is given that $6^{2x+1} = 9 - 3(6^{-2x})$.

(i) With the use of substitution, show that $6^{2x+1} = 9 - 3(6^{-2x})$ can be expressed as a quadratic equation. [2]

(ii) Hence, solve the equation to find the values of x . [3]

- 3 (a) By writing the general term in the binomial expansion of $\left(\frac{4}{5}x^3 - \frac{1}{x^5}\right)^7$ or otherwise, explain why every term is dependent on x .

- (b) In the expansion of $(1+5x)^n$, it is given that the ratio of the coefficients of x^2 and x^3 is 3:25. Find the value of n . [4]

4 (i) Without the use of a calculator, show that $\cos 225^\circ - \sin 75^\circ = \frac{-3\sqrt{2} - \sqrt{6}}{4}$.

- (b) Hence, without using a calculator, find the values of the real numbers a and b if

[3]

$$\cos 225^\circ - \sin 75^\circ = (a+1)\sqrt{2} - b\sqrt{6}.$$

- 5 A simple model for the tide height (h) in metres, at a harbour is given to be $h = a \sin \frac{t}{b} + c$, where t is time in hours measured from midnight.

The mean sea level at the harbour is 2 m and the time between successive high tides is 12 hours. At 0300 h, there is a high tide of 4 m and at 0900 h, there is a low tide of 0 m.

- (i) Find the values of a , b and c .

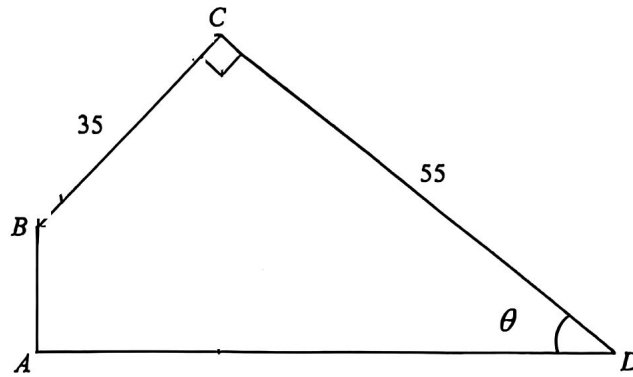
[3]

It is considered safe for ships to leave the harbour when the tide is at least 3.5 m high.

- (ii) Find the total duration, in hours and minutes, during one full tidal period for which it is safe for ships to leave the harbour.

[3]

- 6 A factory was built on a piece of land $ABCD$ as shown below. It is given that $BC = 35$ m, $CD = 55$ m and $\angle ADC = \theta$.

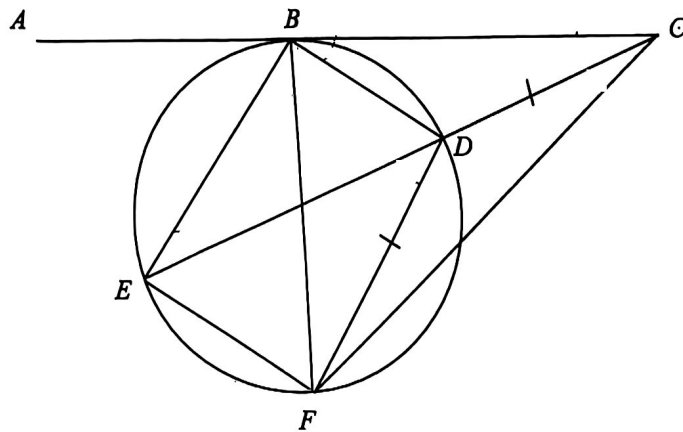


- (i) Show that the perimeter (P) of the land is given by $P = 90 + 20 \cos \theta + 90 \sin \theta$. [3]

- (ii) Express P in the form $90 + R \cos(\theta - \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$. [3]

- (iii) Find the maximum value of P and the corresponding value of θ .

-
- 7 The diagram shows a circle where ABC is a tangent to the circle at point B . CDE and CF are straight lines, $FD = DC$ and B, D, F and E are points on the circle.



- (i) Show that triangle BEC is similar to triangle DBC .

[2]

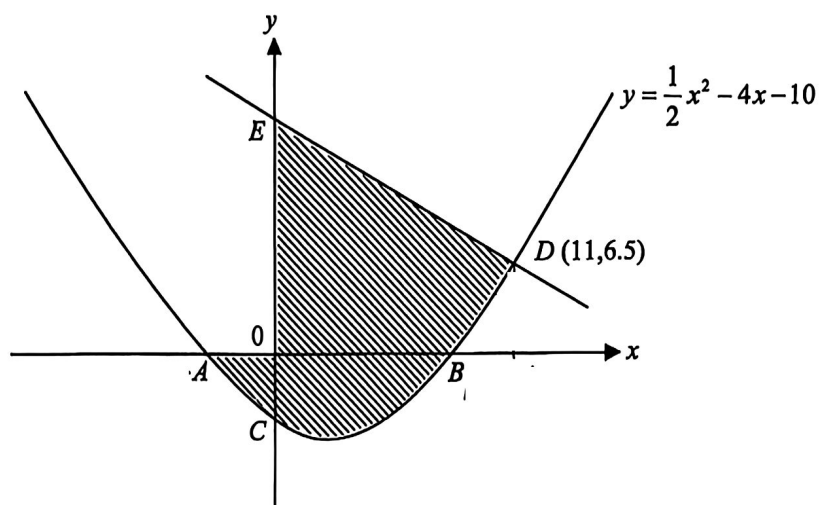
(ii) Show that $FD \times CE = CB^2$.

[2]

(iii) Prove that $\angle EBF = 2 \times \angle DCF$.

[3]

- 8 The diagram shows part of the curve $y = \frac{1}{2}x^2 - 4x - 10$ intersecting the x -axis at A and B and the y -axis at C . The normal to the curve at point $D(11, 6.5)$ meets the y -axis at E .



- (i) Find the equation of the normal DE .

[3]

(ii) Find the area of the shaded region.

[6]

- 9 It is known that m and n are related by the equation $n = p(2^{mq})$, where p and q are constants.

m	1	2	3	4	5
n	79.998	639.88	5119.1	40954	327642

- (i) Using the grid provided on Pg 17, plot $\lg n$ against m and draw a straight-line graph to illustrate the information.

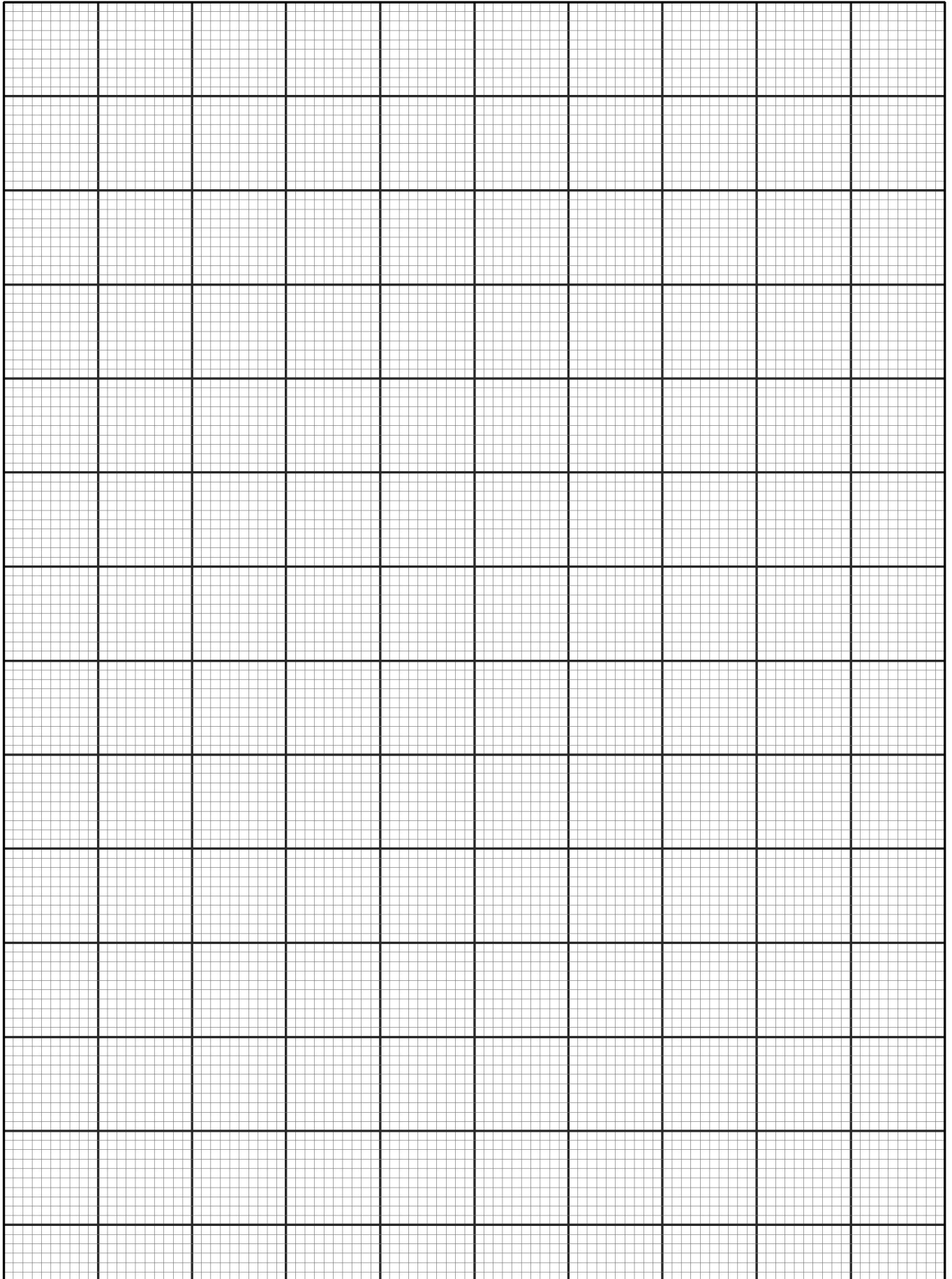
[3]

- (ii) Use your graph to estimate the values of p and q .

[4]

- (iii) By drawing a suitable straight line on the same grid on Pg 17, solve the equation $10^{2-m} = p(2^{mq})$.

[2]



- 10 (a) Particle A moves along a straight line and its acceleration is given by $a = 5e^{3-2t} - 10$, where t is the time in seconds after passing a fixed point O . At the start, its velocity is 10 m/s.

(i) Find an expression for velocity, in terms of e .

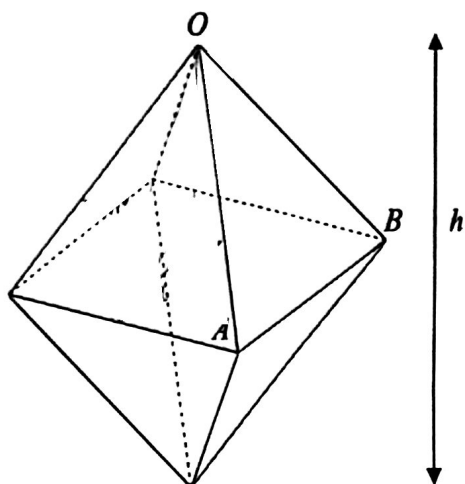
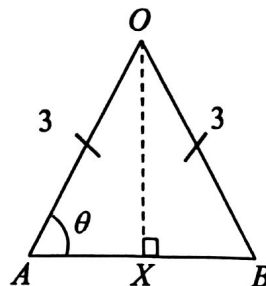
[2]

(ii) Calculate the maximum velocity reached.

[4]

- (b) Particle B moves along a straight line and its velocity is given by $v = 20 \sin t - 10$, where t is the time in seconds after passing a fixed point O .
- (i) Given $t < 5$, find the values of t when B is at instantaneous rest. [2]
- (ii) Find B 's average speed for the first 4 seconds. [4]

- 11 Diagram *A* shows a square bipyramid made up of 8 congruent triangles. The total height is h cm. Diagram *B* shows one of the isosceles triangular faces OAB with dimensions (in cm) given.

Diagram *A*Diagram *B*

- (i) In Diagram *B*, OX is perpendicular to AB . Express AB and OX in terms of θ .

[2]

- (ii) Hence, by expressing h in terms of θ , show that the total volume of the square bipyramid, V , is given by $V = 72 \cos^2 \theta \sqrt{-\cos 2\theta}$.

[3]

(iii) Find the value of θ that gives the square bipyramid a maximum volume.

[7]